

CO3 – AT1 – Debugging

Name: Kandath Haneesh

QUESTIONS

Q1
Try with resources Mouse
25 marks

Q2
Try-Catch-Finally Basics
25 marks

Q3
wait() and notify() Coordination
25 marks

Q4
Basic Thread Creation and Start
25 marks

4/4 answered

Q2 Try-Catch-Finally Basics25 marks

```
public class TryCatchFinallyDemo {
    public static void main(String[] args) {
        int[] numbers = {1, 2, 3};

        try {
            System.out.println(numbers[5]); // ArrayIndexOutOfBoundsException expected here
        } catch (NullPointerException e) { // Bug 1: wrong exception type caught, real exception not handled
            System.out.println("Caught null pointer");
        }

        try {
            int value = Integer.parseInt("abc");
        } catch (Exception e) {
        }

        String str = null;
        try {
            System.out.println(str.length());
        } catch (NullPointerException e) {
            return;
        } finally {
            System.out.println("Cleanup");
            str.length();
        }

        try {
            riskyMethod();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    static void riskyMethod() throws java.io.IOException {
        throw new java.io.IOException("failure");
    }
}
```

Your Code

```
1 public class TryCatchFinallyDemo {
2     public static void main(String[] args) {
3         int[] numbers = {1, 2, 3};
4         try {
5             System.out.println(numbers[5]);
6         } catch (ArrayIndexOutOfBoundsException e) {
7             System.out.println("Caught array index error");
8         }
9         try {
10            int value = Integer.parseInt("abc");
11            System.out.println(value);
12        } catch (NumberFormatException e) {
13            System.out.println("Invalid number");
14        }
15    }
16 }
```

Input

stdin input

Output

Caught array index error
Invalid number
Caught null pointer
Cleanup
IOException caught

Reg No: 192324084

QUESTIONS

Q1
try-with-resources Misuse
25 marks

Q2
Try Catch Finally Basics
25 marks

Q3
wait() and notify() Coordination
25 marks

Q4
New Thread Creation and Start
25 marks

4/4 answered

Q1 try-with-resources Misuse

25 marks

```
import java.io.*;

public class TryWithResourcesDemo {
    public static void main(String[] args) {
        BufferedReader reader = new BufferedReader(new FileReader("data.txt"));
        try (reader) {
            String line = reader.readLine();
            System.out.println(line);
        } catch (IOException e) {
            e.printStackTrace();
        }

        try (FileWriter writer = new FileWriter("out.txt")) {
            writer.write("Hello");
        }

        class NotCloseable {
            void write() {}
        }

        try (NotCloseable nc = new NotCloseable()) {
            nc.write();
        }

        try (FileReader fr1 = new FileReader("a.txt");
            FileReader fr2 = new FileReader("b.txt")) {
            {
                fr1.close();
                fr2.close(); close (double-close, though usually harmless it's a logic bug/anti-pattern here compounded by bug 5)
            } catch (IOException e) {
            } finally {
                fr1.close();
            }
        }
    }
}
```

Your Code

```
1 import java.io.*;
2 public class TryWithResourcesDemo {
3     public static void main(String[] args) {
4         try (BufferedReader reader = new BufferedReader(new FileReader("data.txt"))) {
5             String line = reader.readLine();
6             System.out.println(line);
7         } catch (IOException e) {
8             e.printStackTrace();
9         }
10
11         try (FileWriter writer = new FileWriter("out.txt")) {
12             writer.write("Hello");
13         } catch (IOException e) {
14             e.printStackTrace();
15         }
16
17         try (FileReader fr1 = new FileReader("a.txt");
18             FileReader fr2 = new FileReader("b.txt")) {
19             {
20                 fr1.close();
21                 fr2.close(); close (double-close, though usually harmless it's a logic bug/anti-pattern here compounded by bug 5)
22             } catch (IOException e) {
23             } finally {
24                 fr1.close();
25             }
26         }
27     }
28 }
```

Input

stdin input

Output

```
java.io.FileNotFoundException:
data.txt (The system cannot
find the file specified)
or
java.base/java.io.FileInputStream
or FileOutputStream
```

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QUESTIONS

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try with resources Mouse
25 marks

Q2

Try Catch Finally Basics
25 marks

Q3

wait() and notify() Coordination
25 marks

Q4

Basic Thread Creation and Start
25 marks

4/4 answered

Q2 Try-Catch-Finally Basics

25 marks

```
public class TryCatchFinallyDemo {  
    public static void main(String[] args) {  
        int[] numbers = {1, 2, 3};  
  
        try {  
            System.out.println(numbers[5]); // ArrayIndexOutOfBoundsException expected here  
        } catch (NullPointerException e) { // Bug 1: wrong exception type caught, real exception not handled  
            System.out.println("Caught null pointer");  
        }  
  
        try {  
            int value = Integer.parseInt("abc");  
        } catch (Exception e) {  
        }  
  
        String str = null;  
        try {  
            System.out.println(str.length());  
        } catch (NullPointerException e) {  
            return;  
        } finally {  
            System.out.println("Cleanup");  
            str.length();  
        }  
  
        try {  
            riskyMethod();  
        } catch (IOException e) {  
            e.printStackTrace();  
        }  
    }  
  
    static void riskyMethod() throws java.io.IOException {  
        throw new java.io.IOException("failure");  
    }  
}
```

Your Code

```
1 public class TryCatchFinallyDemo {  
2     public static void main(String[] args) {  
3         int[] numbers = {1, 2, 3};  
4         try {  
5             System.out.println(numbers[5]);  
6         } catch (ArrayIndexOutOfBoundsException e) {  
7             System.out.println("Caught array index error");  
8         }  
9         try {  
10            int value = Integer.parseInt("abc");  
11            System.out.println(value);  
12        } catch (NumberFormatException e) {  
13            System.out.println("Invalid number");  
14        }  
15    }  
16 }
```

Input

stdin input

Output

```
Caught array index error  
Invalid number  
Caught null pointer  
Cleanup  
IOException caught
```

25 marks

Q4

Basic Thread Creation and Start
25 marks

4/4 answered

```
lock.notify();
}
}

public void consume() {
    if (!dataReady) {
        try {
            lock.wait();
        } catch (InterruptedException e) {
        }
    }
    System.out.println("Data consumed");
}

public void consumeCorrectly() {
    synchronized (lock) {
        if (!dataReady) {
            try {
                lock.wait();
            } catch (InterruptedException e) {
                Thread.interrupted();
            }
        }
    }
}

public static void main(String[] args) {
    WaitNotifyDemo demo = new WaitNotifyDemo();
    Thread producer = new Thread(demo::produce);
    Thread consumer = new Thread(demo::consumeCorrectly);
    consumer.start();
    producer.start();
    consumer.join();
}
}
```

Your Code

```
1 public class WaitNotifyDemo {
2     private boolean dataReady = false;
3     private final Object lock = new Object();
4     public void produce() {
5         synchronized (lock) {
6             dataReady = true;
7             System.out.println("Data produced");
8             lock.notify();
9         }
10    }
11    public void consume() {
12        synchronized (lock) {
13            while (!dataReady) {
14                try {
15                    lock.wait();
16                } catch (InterruptedException e) {
```

Input

stdin input

Output

Data produced
Data consumed

QUESTIONS

Q1
try with resources Mutex
25 marks

Q2
try Catch Finally Basics
25 marks

Q3
wait() and notify() Coordination
25 marks

Q4
Basic Thread Creation and Start
25 marks

4/4 answered

Q4 Basic Thread Creation and Start

25 marks

```
public class ThreadBasics {  
    public static void main(String[] args) {  
        Thread t1 = new Thread(new MyTask());  
        t1.run();  
        MyTask task = new MyTask();  
        task.start();  
  
        Thread t2 = new Thread(task);  
        t2.start();  
        t2.start();  
  
        System.out.println(t1.getName());  
  
        try {  
            t2.join()  
        } catch (InterruptedException e) {  
        }  
    }  
  
    class MyTask implements Runnable {  
        public void run() {  
            System.out.println("Task running on: " + Thread.currentThread().getName());  
        }  
    }  
}
```

Your Code

```
1 public class ThreadBasics {  
2     public static void main(String[] args) {  
3         Thread t1 = new Thread(new MyTask());  
4         MyTask task = new MyTask();  
5         Thread t2 = new Thread(task);  
6         t1.start();  
7         t2.start();  
8         try {  
9             t1.join();  
10            t2.join();  
11        } catch (InterruptedException e) {  
12            Thread.currentThread().interrupt();  
13        }  
14    }  
15 }  
16 class MyTask implements Runnable {  
17     public void run() {  
18         System.out.println("Task running on: " + Thread.currentThread().getName());  
19     }  
20 }
```

Input

stdin input

Output

```
Task running on: Thread-0  
Task running on: Thread-1
```

Run

Save